



Integrating deep learning with physics-based model for predicting grassfire spread

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Abstract Shrublands and grasslands, which constitute approximately 70% of Australia's vegetation, play a critical role in global wildfire-prone regions. To advance the understanding of grass fire spread, a three-dimensional, physics-based fire model provides valuable insights into fire dynamics. However, such models are computationally intensive and time-consuming. To address these challenges, we constructed an extensive numerical database comprising 64,000 high-fidelity wildfire simulation cases and implemented a Long Short-Term Memory neural network architecture. The model demonstrates strong predictive performance,

achieving a coefficient of determination (R^2) of 0.96 on training data, indicating excellent agreement with the physics-based simulation outputs. By utilizing coordinates from five reference points to predict fire front movement, this approach offers a novel method for analysing fire dynamics in homogeneous fuel beds with an average deviation of less than 2.5%. Combining the strengths of physics-based modelling and deep learning, our research enhances fire spread prediction accuracy of over 95% while significantly reducing computational demands. Future efforts will focus on refining the model, expanding the dataset, and incorporating additional variables to improve predictive capabilities and operational applicability.

Rahul Wadhvani and Xiaoning Zhang contributed equally and both should be considered as the first author.

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Introduction

Wildfire propagation is a complex phenomenon influenced by many interconnected factors, including the type of ignition source, the composition and characteristics of the fuel, ambient weather conditions, and topographical features (Sullivan 2017a, 2017b). Substantial advancements have

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been made in the science of fire behaviour prediction across various regions. However, even in countries with robust expertise in fire behaviour research and operational fire management, encompassing situational awareness, fire spread forecasting, and wildfire disaster policies, unexpected fire events challenge emergency response agencies, sometimes resulting in fatalities.

A primary factor contributing to these unexpected fire events is the dynamic behaviour of fire. Most commonly employed operational fire models rely on empirical correlations that link fire spread rates to ambient parameters and fuel characteristics, operating under the quasi-steady state assumption. This assumption posits that a specific set of ambient parameters and fuel properties can uniquely determine the rate of spread, intensity, flame height and overall fire behaviour. These empirical correlations generally are derived from controlled experimental studies conducted in relatively benign weather conditions, characterised by low to moderate wind speeds and ambient temperatures around 30 °C. Consequently, these models are often inadequate for accurately representing complex weather scenarios, resulting in diminished predictive accuracy during severe fire events (Lareau and Clements 2016, 2017; Liu et al. 2021).

Dynamic fire behaviours, including vorticity-driven lateral flow, fire whirls, pyrogenic winds, and spotting phenomena, often violate the quasi-steady state assumption. These behaviours can significantly alter fire dynamics without corresponding changes in ambient parameters. Accurately modelling such dynamic fire behaviour necessitates detailed information regarding local wind patterns, heat loads (both convective and radiative), and vegetative fuel classification, changing over time and space. Unfortunately, this level of granularity exceeds the current capabilities of operational fire models (mainly carried out on > 30 m scale) (Finney 1998; Tolhurst et al. 2008). These models predominantly rely on lab-scale correlations that are not designed to accommodate the complexities inherent in these scenarios.

A three-dimensional physics-based fire model represents one potential approach for studying the dynamic behaviour of fire. However, these models demand significant computational resources, extended computational times, and rigorous verification and validation against experimental data (Bakhshaii and Johnson 2019; Morvan 2020; Ronchi et al. 2017; Sullivan 2009). Additionally, these models are inherently deterministic, meaning that their estimations and predictions rely heavily on static initial conditions, such as initial fire location, fuel type and orientation, and terrain features, which limits their applicability to real-world scenarios. The capability of these fire models is generally limited to simulating wildfires over distances of only a few hundred meters in a reasonable timeframe (Ronchi et al. 2019). As a result, researchers must navigate a trade-off between modelling accuracy and computational cost (Singh et al. 2024).

Recently, there has been a paradigm shift towards utilising artificial intelligence (AI) techniques for simulating fire behaviour (Huang et al. 2022; Naser 2021), with applications extending into the field of wildfire science (Jain et al. 2020). Various studies have been reviewed focusing on the application of machine learning in wildfire science (Jain et al. 2020). It observed that most studies focused on its utilisation in the domain of fire management and the decision-making aspect of wildfire. The application of machine learning and deep learning techniques for simulating and understanding fire behaviours is minimal, including the ANN-based fire behaviour predictions (Chetehouna et al. 2015; Vakalis et al. 2004) and estimation of optimised parameters (Denham et al. 2012; Lautenberger et al. 2006). Integrating deep learning with a wildfire model to estimate wildfire behaviour, and the rate of fire spread is still nascent (Khanmohammadi et al. 2022).

However, one work (Hodges and Lattimer 2019) integrated a deep learning (deep convolutional inverse graphics network) model with the FARSITE fire model, which utilises Rothermel's empirical correlation (Rothermel 1972) to simulate fire behaviour. They observed that a deep learning-based model could predict fire burn maps of similar accuracy as estimated using FARSITE and significantly reduced computation time for heterogeneous spatial conditions. Some researchers (Allaire et al. 2021) also carried out a similar approach of integrating a deep learning-based model with the ForeFire fire model, which also uses Rothermel's empirical model to simulate fire behaviour. They observed that their learning model has a mean absolute error of ~ 35% in estimating the burnt area. Additionally, integrating physics with AI into wildfire modeling demonstrates a good understanding of enhancing the wildfire modeling accuracy (Dabrowski et al. 2023).

In conclusion, adopting deep learning models to forecast wildfires within an acceptable timeframe accurately is feasible. However, this approach heavily relies on the quality of the underlying database, as the model learns from and fits the data correlations present in the database, which are often based on empirical correlations used in traditional modeling software. These integration methods implicitly depend on the empirical relationships established for fire behaviour modelling, which can introduce similar issues and uncertainties into deep-learning integrated fire models (Valero et al. 2021). Additionally, empirical correlations for estimating fire behaviours have not been developed for all vegetation types worldwide; typically, well-known empirical models or their modified forms are employed in such cases. The efficacy of these models in predicting the rate of fire spread and burned area is critical (Allaire et al. 2021; Cruz and Alexander 2013; Li et al. 2024). Therefore, to enhance the effectiveness of deep learning models in wildfire forecasting, it is essential to improve the reliability of the data used.

Physics-based fire models offer a powerful alternative to empirical correlations by explicitly resolving turbulence and combustion dynamics, thereby avoiding the simplifications inherent in empirical approaches (Dabrowski et al. 2023; Vanella et al. 2021). However, their operational use is often restricted due to significant computational resource requirements (K McGrattan et al. 2022a, b; Sullivan 2009). This limitation motivates our development of a hybrid approach that combines deep learning with high-fidelity physics-based simulations. By leveraging synthetic data derived from computational fluid dynamics rather than relying on empirical parameterizations, our methodology captures critical transient phenomena, including wind-fuel interactions and vortex-driven flank dynamics, that conventional empirical models typically fail to resolve. While empirical correlations perform adequately under steady-state conditions, their predictive capability degrades significantly during dynamic fire events, precisely where our physics-informed approach demonstrates superior performance.

This integration has been successfully implemented in confined space fires, such as tunnels (Zhang et al. 2022) and buildings (Hodges et al. 2019; Zeng et al. 2022). The present study discusses preliminary efforts to combine a deep learning model with a three-dimensional physics-based fire model to estimate fire behaviour by predicting the rate of fire spread. To accomplish this, a numerical database of wildfire simulations using the physics-based fire model was established, which later served as a synthetic dataset for training the deep learning model. The integrated model is cross-validated using independent numerical cases and experimental data to evaluate its efficacy.

Physics-based grassfire simulation

Grassfire model evolution

Shrublands and grasslands cover approximately 70% of Australian vegetation and have a significant presence worldwide in wildfire-prone regions (Jackson et al. 2017). Significant experimental studies (Cruz et al. 2017) have been conducted to develop a correlation to estimate the rate of fire spread. McArthur model (McArthur 1966) has been widely used to compute the rate of fire spread (rate of spread in km/h) for Australian grasslands and is represented by Eq. 1,

$$ROS = C_1 f(C, T, H, U_{10}, W, MC) \quad (1)$$

where, C_1 is empirical constant, C is the curing index (which represents the proportion of dead fuel in total fuel load), T is the ambient temperature ($^{\circ}\text{C}$), H is the relative humidity in per cent, U_{10} is the open wind speed at 10 m height (km/h), W is the fuel load (tonnes/ha), and MC is moisture content.

Cheney et al. (Cheney et al. 1993) and Cruz et al. (Miguel G. Cruz et al. 2017) also proposed a similar correlation incorporating additional parameters to estimate rate of spread (ROS) in grasslands. The relative impact of each parameter on the rate of spread is different. Cruz et al. (M.G. Cruz et al. 2015) observed that the ROS in grassland and shrublands have a significantly higher dependency on the fuel moisture content of vegetative fuel (including dead-fuel moisture content), slope of the vegetation, near-surface wind speed, and fuel load (including height of the vegetation). Various other researchers also have confirmed the above observations (Bartoli et al. 2011; Cheney and Sullivan 2008; Coen 2018; Liu et al. 2021; Rothermel and Deeming 1980; Sullivan 2017a, 2017b).

Numerical techniques

Grassfires have been simulated using a physics-based fire model by various researchers due to their simplicity and homogeneity (Hilton et al. 2016; Mell et al. 2013, 2007; Moinuddin et al. 2018; Sullivan 2007; Vanella et al. 2021). Thus, in the present study, the effects of only three parameters- open wind speed, moisture content of vegetative fuel, and height of the vegetative fuel bed, were considered to develop a synthetic dataset that will be used for a machine learning-based model.

A set of grassfire simulations was carried out using Fire Dynamics Simulator (FDS) (ver. 6.7.9). The experimental grassfire observations were recorded in July and August 1986 in the Northern Territory of Australia by Cheney et al. 1993 and have been commonly referred as C064 and F19 case. The C064 case corresponds to 100×100 m grassfire in *Eriachne burkittii* (commonly called kerosene grass), and the F19 case corresponds to 200×200 m grassfire in *Themeda australis* (commonly called kangaroo grass) (Cheney et al. 1993). Earlier studies (Mell et al. 2007; Moinuddin et al. 2018) carried out validation with the experimental data using an older extension model of FDS, and Vanella et al. 2021 have validated with an earlier version of FDS (ver. 6.7.0) to simulate grassfires and compared the efficacy of different inbuilt models. Vanella et al. 2021 showed the efficacy in estimating the head of the fire front using the boundary fuel model of FDS was better than using the Lagrangian particle model and was much better than the Level-set method.

In the boundary fuel model, fuel is represented as a porous solid layer with a thickness equal to fuel height. The surface vegetation layer is too shallow to be resolved explicitly, as is done when using the Lagrangian particle model to represent the vegetation. The eddy inside the porous media is not explicitly resolved; thus, the convective mode of heat transport is very limited in the porous media and is mainly dominated by the radiative mode of transport. Furthermore,

the height of vegetative fuel is also assumed to be unresolved on the grid. FDS Technical guide (2022b) and (Vanella et al. 2021) have provided the governing equations of the boundary fuel model used in the present study.

Utilising previous research work, the present work utilises a similar simulation domain setup, which was used in the validation of the F19 case. A rectangular simulation domain was considered with dimensions of $240 \times 240 \times 20$ m (spanning from -20 to 220 in longitudinal (X –) direction, -120 to 120 in lateral (Y –) direction, and 0 – 20 in height (Z – direction)) as shown in Fig. 1. The combustible vegetative fuel comprised of Kangaroo grass of 200×200 m (spanning from 0 to 200 in longitudinal (X –) direction and -100 to 100 in lateral (Y –) direction) was considered as shown in Fig. 1.

Rectilinear grid $\Delta x = \Delta y = \Delta z = 0.5$ m was assumed based on previous validation studies carried out by Vanella et al. 2021. Atmospheric boundary layer profile wind is prescribed at the inlet of a domain at $X = -20$ m which is denoted as U in Fig. 1 and is defined by Eq. 2.

Monin–Obukhov similarity theory was used to prescribe the wind profile inlet of domain based on surface and atmospheric condition (further details are available in FDS user and technical guides (Kevin McGrattan et al. 2022a, b)).

$$U(z) = \frac{u_*}{\kappa} \left[\ln \left(\frac{z}{z_0} \right) - \psi_m \left(\frac{z}{L} \right) \right] \quad (2)$$

$$\text{where, } \psi_m \left(\frac{z}{L} \right) = \begin{cases} -5 \frac{z}{L}, & L \geq 0 \\ 2 \ln \left[\frac{1+\zeta}{2} \right] + \ln \left[\frac{1+\zeta^2}{2} \right] - 2 \tan^{-1}(\zeta) + \frac{\pi}{2}, & L < 0 \end{cases}$$

$$\text{and } \zeta = \left(1 - \frac{16z}{L} \right)^{\frac{1}{4}}.$$

where, u_* is the friction velocity, $\kappa = 0.41$ is the Von Kármán constant, L is the Obukhov length, z_0 is the aerodynamic roughness length (for the present study, it is assumed to be 0.3 as grasslands were simulated), z is the height above the ground.

Table 1 presents the fuel properties utilized in this study, including four discrete values for each of three key parameters: wind speed at 2 m height (U_2), grass height,

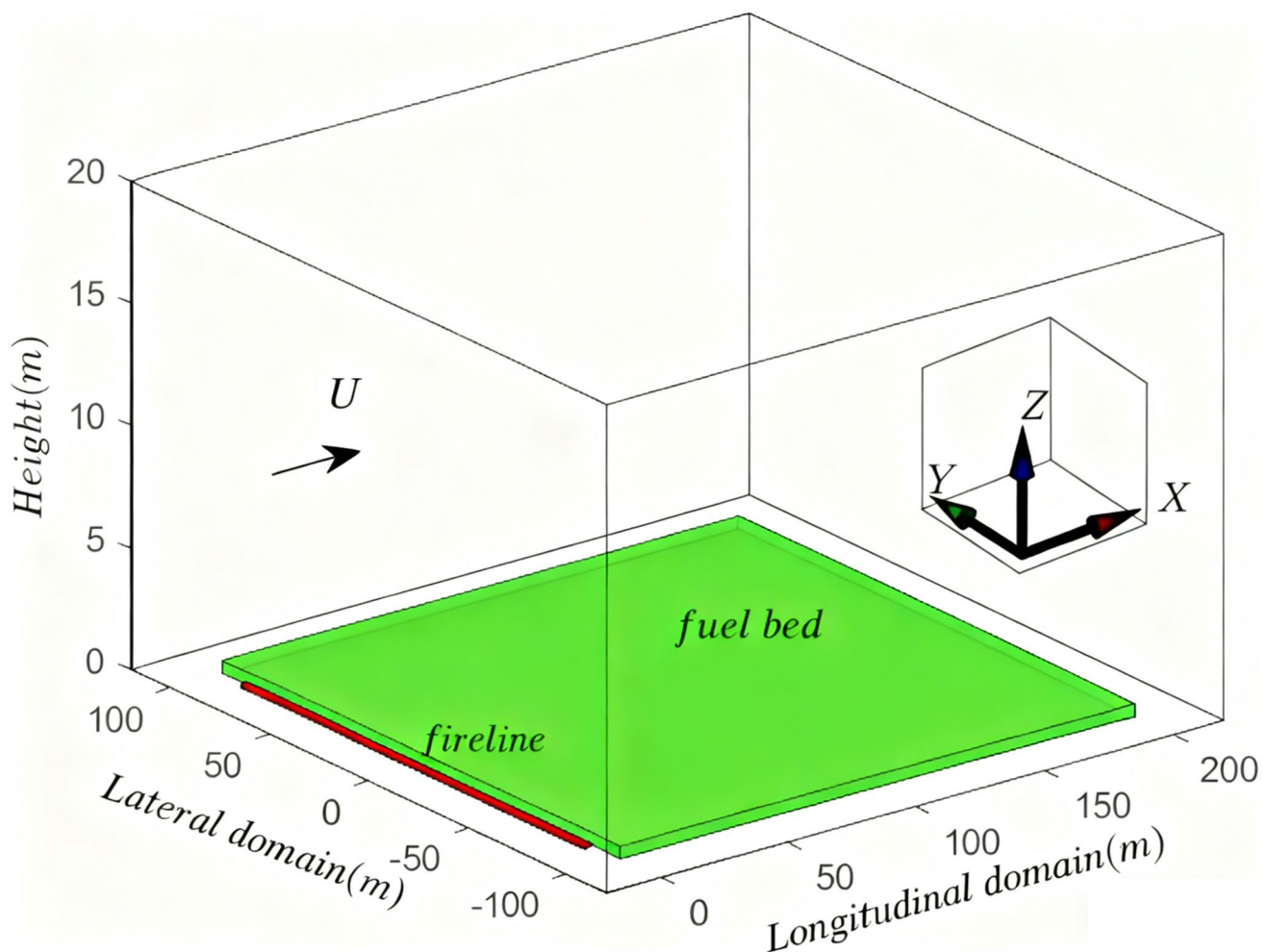


Fig. 1 Schematic layout of modelling grassfire via computational fluid dynamics (CFD)

Table 1 Fuel properties used in the present study

Fuel properties	Value	Reference
Vegetative fuel	<i>T. australis</i> (Kangaroo grass)	Cheney et al. (1993)
Ambient temperature	34 °C	
Surface-to-volume ratio	12,240 m ⁻¹	
Bulk mass per unit area	0.313 kg/m ²	
Chemical composition	$C_6H_{10}O_5$	Vanella et al. (2021)
Heat of combustion	15,600 kJ/kg	Mell et al. (2007)
Soot yield	0.015 kg/kg	
Char Yield	0.2 kg/kg	
Specific heat capacity	1.5 kJ/kg.K	
Conductivity	0.1 W/m.K	
Density	512 kg/m ³	
Heat of pyrolysis	418 kJ/kg	
Obukhov length	− 500 m	Vanella et al. (2021)
Aerodynamic roughness length	0.03 m	
Drag coefficient	2.8	
A_{pyr} (pyrolysis)	36,300 s ⁻¹	Kevin McGrattan et al. (2022a, b)
E_{pyr} (pyrolysis)	60,300 J/mol	
A_{dry} (drying)	600,000 K ^{0.5} s ⁻¹	
E_{dry} (drying)	48,200 J/mol	
A_{char} (pyrolysis)	430 m/s	
E_{char} (pyrolysis)	74,800 J/mol	
Grass height (h_{grass})	420, 480, 540, 600 mm	Considered variables in the present study
Wind speed at 2 m height (U_2)	3, 4, 5, 6 m/s	
Moisture content (MC)	4, 5, 6, 7%	

and vegetative fuel moisture content. These parameters were selected for developing a synthetic dataset to train the machine learning model. The parameter selection was informed by M.G. Cruz et al. (2015), whose empirical correlations for rate of spread in grassfire models served as our foundation.

The study focuses on undisturbed natural grassland with a representative height of 50 cm, consistent with kangaroo grass vegetation as examined in the F19 experiment. Wind speed and moisture content variations were determined based on typical field conditions for tallgrass vegetation, enabling the development of robust empirical correlations. Our scenarios span the complete range of MK4 grass-fire danger levels, from moderate to extreme conditions, approaching catastrophic thresholds in some cases (M.G. Cruz et al. 2015, Cruz et al. 2017). To maintain consistency with the F19 experimental framework, we assumed homogeneous surface fuel distribution throughout the analysis.

$$MC = \left[\frac{97.7 + 4.06RH}{T + 6} - 0.00845 \times RH + \frac{3000}{C} - 30 \right] \quad (3)$$

Moisture content (MC) of the fuel bed is dependent on ambient relative humidity (RH) and temperature (T), and the curing index of grassland (C) as correlated by Eq. 3 (McArthur 1966), where, T is used in °C, RH in %, and C in %.

Numerical analysis

The numerical simulations were carried out for a set of 64 scenarios for the variables considered in the present study (see Table 1). Figure 2a illustrates a typical snapshot view of fire progression for one of the cases. The fireline is divided at five cross-section positions at $y = 0, \pm 30, \pm 60$ m to measure fire front location and the spread rate along those lines, represented by lines A, B, C, D, and E. Figure 2b presents the fire line positions at different wind speeds for 0.42 m grass height and 4% moisture content. Figure 2c, d reveals the point positions of A and E during the spread process. It is noteworthy that neither the head nor the flanks can withstand 3-m wind speeds under conditions of 0.42 m grass height and 4% moisture content. The details of the firefront location for each measurement line under different circumstances (A, B, C, D, and E) are provided in Supplementary Figs. S1–S5.

Our findings indicate a positive correlation between wind speed and flame propagation. Specifically, higher wind speeds facilitate a more rapid advancement of the flame front, resulting in shorter times for the fire to reach its endpoints, as illustrated in Fig. 2c, 2d. This relationship highlights the crucial role of environmental conditions, particularly wind, in shaping fire dynamics. The trend is further supported by the increasing rate of spread curve depicted in

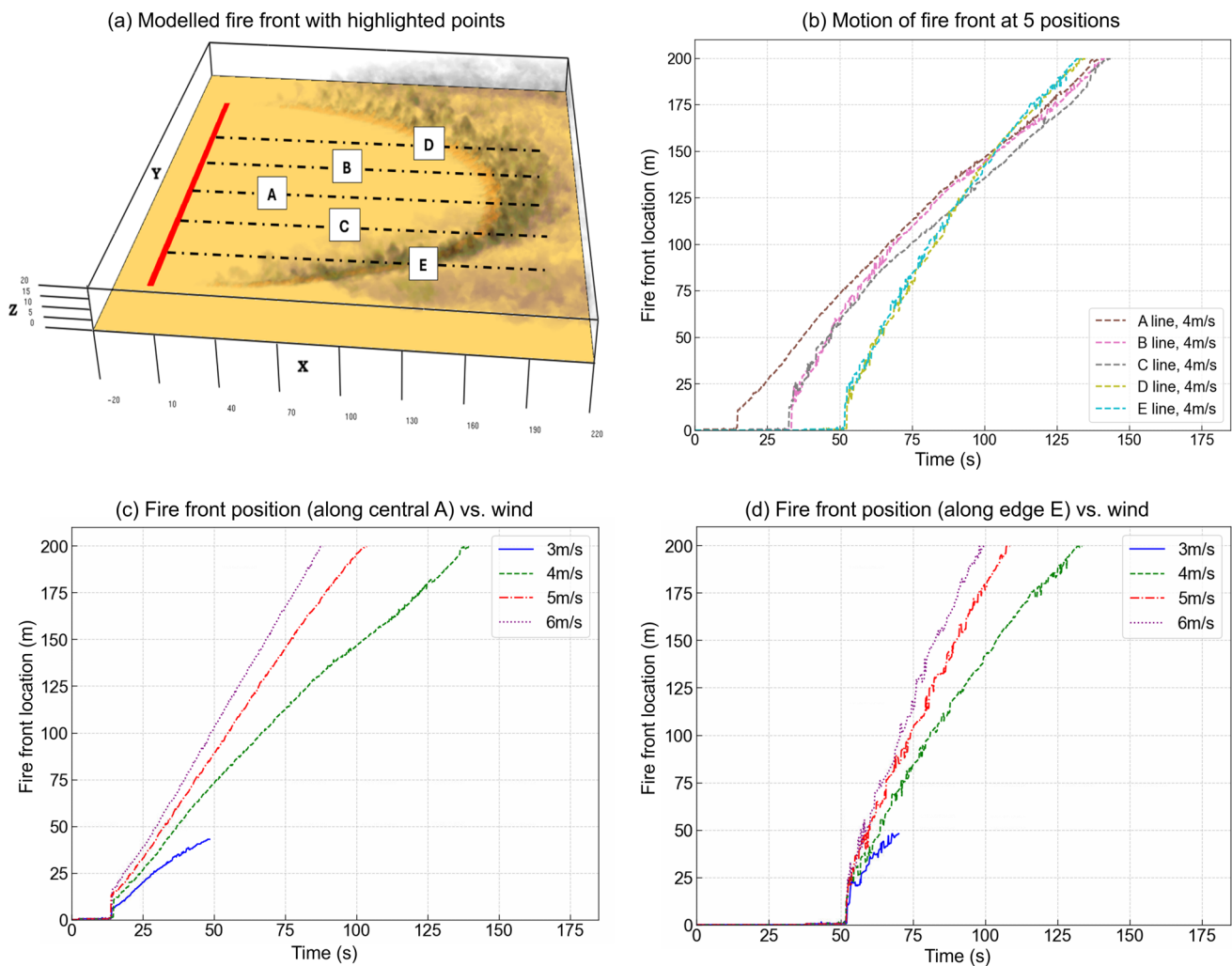


Fig. 2 Snapshot view of fire progression of numerical simulation, **a** multiple measuring locations across the fireline marked by A, B, C, D, and E at $y=0, \pm 30, \pm 60$ m, respectively, **b** fire head rate of spread at different wind speeds for 0.42 m grass height and 4% moisture content, **c** Fire front locations with time for the centreline position A, i.e.

$y=0$ m at different wind speeds for 0.42 m grass height and 4% moisture content, and **d** Fire front locations with time for the centreline position D, i.e. $y=-60$ m at different wind speeds for 0.42 m grass height and 4% moisture content

Fig. 3a. However, we observed that fire propagation is unsustainable at a wind speed of 3 m/s, as the fire extinguishes itself due to insufficient heat flux to sustain its spread in Fig. 3a.

This is a limitation of FDS' boundary fuel method (Innocent et al. 2023). Fire propagation is primarily buoyancy-dominated at lower wind speeds, whereas it becomes shear-dominated at higher wind speeds. This transition is evident in the rate of spread, represented by the slope of the line between the fire front location and time, aligning with observations made by Moinuddin et al. (2018) using the Wildland-Urban Interface Fire Dynamics Simulator (WFDS). At a wind speed of 5 m/s, the rate of change in the firefront location decreases, reducing the slope of this line. This phenomenon occurs due to wind recirculation from the

edges, which slows down the propagation of the fire front. Furthermore, based on the numerical results obtained from the FDS, in which distance and time values were extracted, we applied a simplified speed equation to calculate the fire spread rate. The derived results are presented in Fig. 3b. The analysis reveals a clear positive correlation between the rate of fire spread and ambient wind speed, indicating that higher wind speeds consistently correspond to increased fire propagation rates.

This aspect is further highlighted in Fig. 4, which presents cross-sectional views of velocity contour plots. Figure 4 shows the contour of fire propagation for one of the synthetic data sets at a particular time to illustrate the fire behaviour observed. Figure 4a, c plots show flame contour (red), with temperature contours representing the plume

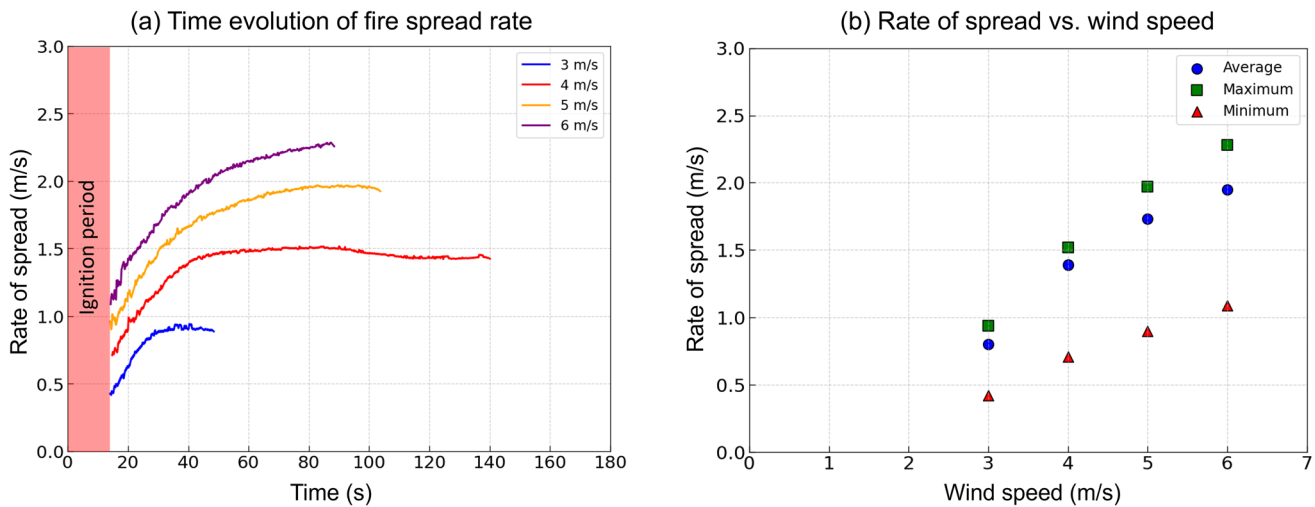


Fig. 3 **a** Fire head rate of spread at different wind speeds for 0.42 m grass height and 4% moisture content, and **b** the correlation between ROS and wind speed for 0.42 m grass height and 4% moisture content

shaded in the yellow background and wind velocity (white arrows) from the top and side view.

The wind velocity vectors are presented to visualise the flow and dynamic behaviour during the grassfire propagation. This dynamic behaviour is developed due to the pressure difference between the firefront's upstream and downstream zones (Eftekharian et al. 2019, 2020). The current snapshot provides insight into instantaneous turbulence observed in the fire progression with a negative value (due to direction) of flow velocity forming inside a solid white line, causing resistance to the fire progression. Figure 4b provides a cross-section view along the YZ plane ahead of the fire front, highlighting two counter-rotating vortices formation in the plume as the fire progresses, similar to the observation made by Shinohara (Shinohara 2021) and Linn et al. (Linn et al. 2012). The detailed description of turbulence behaviour (Jenkins et al. 2007) is not the aim of the present work and will be discussed in future work.

Deep learning methodology

Dataset generation

As discussed previously, numerical simulations were carried out using FDS to generate the synthetic data for the machine learning model. The temperature of the fuel bed is assumed to be representative of the location of the flame front when the temperature of the fuel bed is above 400 °C which has been used in various previous studies using physics-based fire models (Khan et al. 2019; Linn et al. 2002; Mell et al. 2013, 2007; Moinuddin et al. 2018; Morvan et al. 2011; Perez-Ramirez et al. 2017). A set of

64 simulations is generated as a synthetic dataset for the machine learning model. Independent parameters, viz. height of the grass, moisture content of fuel bed, wind speed 2 m above the ground, and simulation time step (1 s), were considered as input parameters as shown in Fig. 5. The input data was generated into a 4 (parameters) $\times$ 1000 (time step) $\times$ 64 (scenario number) matrixes, as shown in the lower left of Fig. 2. Each column means one input parameter and each row represents the input vector of a time step.

To represent the fire front locations in 2D (X, Y) with time, five positions (randomly representatives of Y) are considered to represent the fire front. One head fire location ($Y = 0$) and two flank fire locations on either side ($Y = \pm 30, \pm 60$) were considered. The location of the fire front was considered as output parameters were labelled as (x_{1-5}, y_{1-5}) as highlighted in Fig. 5. As shown in the lower right of Fig. 2, the data matrix ($5 \times 1000 \times 64$) of the model output was constructed. Each row represents one output vector, and different rows mean different time steps, and 1000-time steps are recorded during the 180 s simulation.

A training sample is formulated by combining input and output vectors. One thousand samples were generated for one scenario, and 64,000 samples were obtained for all 64 scenarios in total. After labelling, these samples were scaled between 0 and 1 to reduce the potential influence from the data scale. Before the model training, the dataset was randomly shuffled and divided into the training (80%) and validation datasets (20%), respectively.

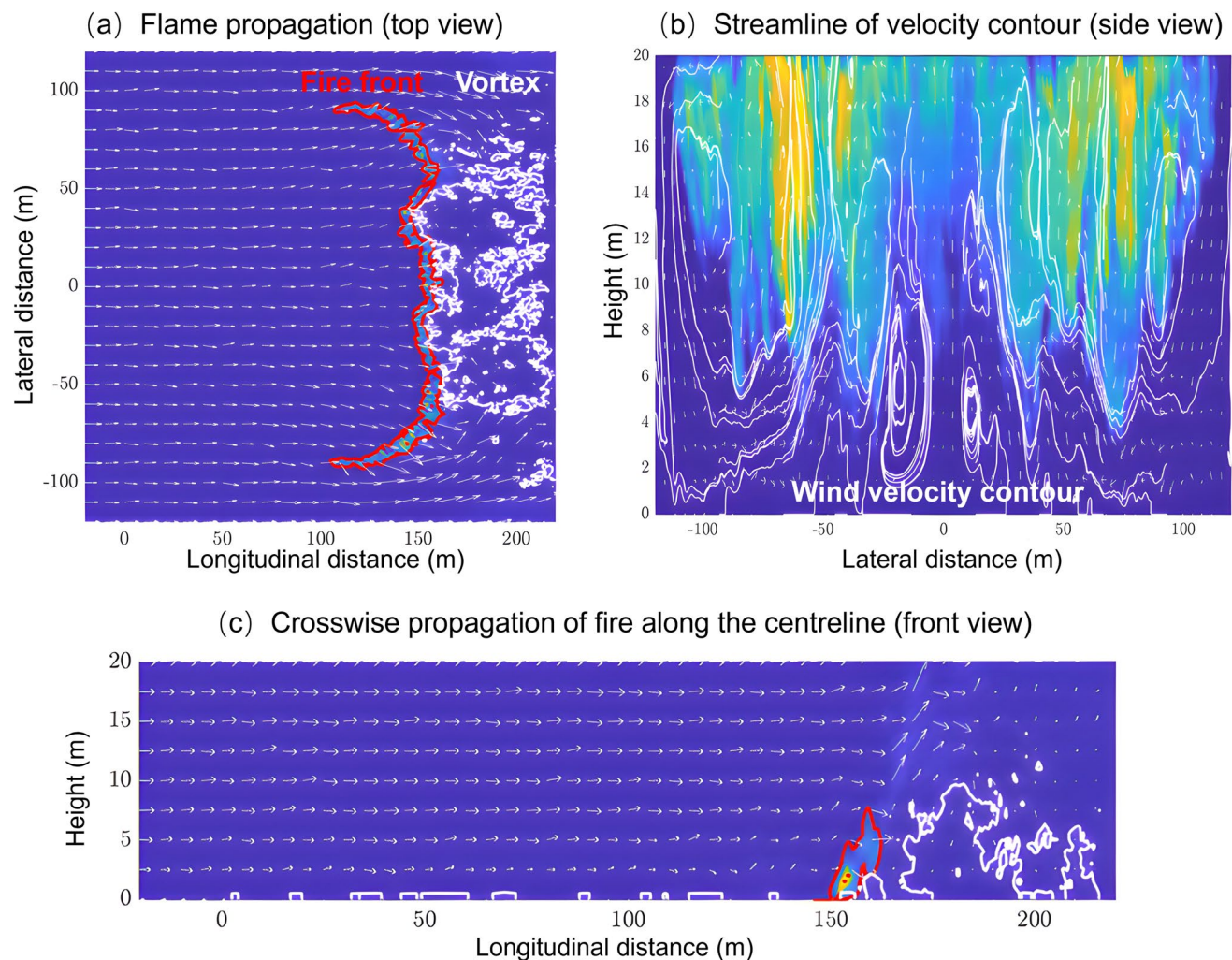


Fig. 4 Velocity contour plotted at the same instant after the start of simulation along the three view angles for the case height of 420 mm, moisture content of 4% and wind speed of 4 m/s. **a** Flame propagation in XY direction highlighting negative velocity formation in front of the head location (represented by solid white lines), **b** Streamline of velocity contour (solid white lines) in YZ direction highlighting

counter-rotating vortices formation ahead of the fire front at $X = 180$ m while the colour represents the flame, and **c** Crosswise propagation of fire at the centreline position ($Y = 0$ m) in the XZ direction while the solid white line represents a zone of negative velocity vector

Model design and training

The Long Short-Term Memory (LSTM) neural network was adopted in the present work, which is designed to extract the features from time-dependent data and has been validated by researchers in the fire scenario (Wu et al. 2022). LSTM networks are ideal for predicting fire line coordinates due to their ability to model long-term dependencies and handle sequential data. Unlike traditional machine learning models, LSTMs excel at capturing temporal patterns in time series data, like historical fire spread, which are critical for accurate predictions. Their gating mechanisms mitigate the vanishing gradient problem, enabling stable learning over long sequences. Additionally, LSTMs adapt to variable-length inputs, making them suitable for irregular wildfire data.

By leveraging these strengths, LSTMs outperform simpler models in capturing complex, nonlinear relationships, providing more reliable fire line forecasts for effective wildfire management.

TensorFlow version 2.7.0 was used to build the proposed LSTM network. The structure of the proposed network is shown in Fig. 6. The input data consists of several different parameters and is formed as a vector with a dimension of 4×1 to facilitate the data reading of code. The proposed model comprises a LSTM layer with 100 states and a subsequent LSTM layer with 10 states. The output layer consists of 5 units to align the dimension of the output vector. A dropout layer with a dropout ratio of 0.1 was selected for each LSTM layer to avoid overfitting. "tanh," "mean squared error (MSE)," and "adam" was chosen as

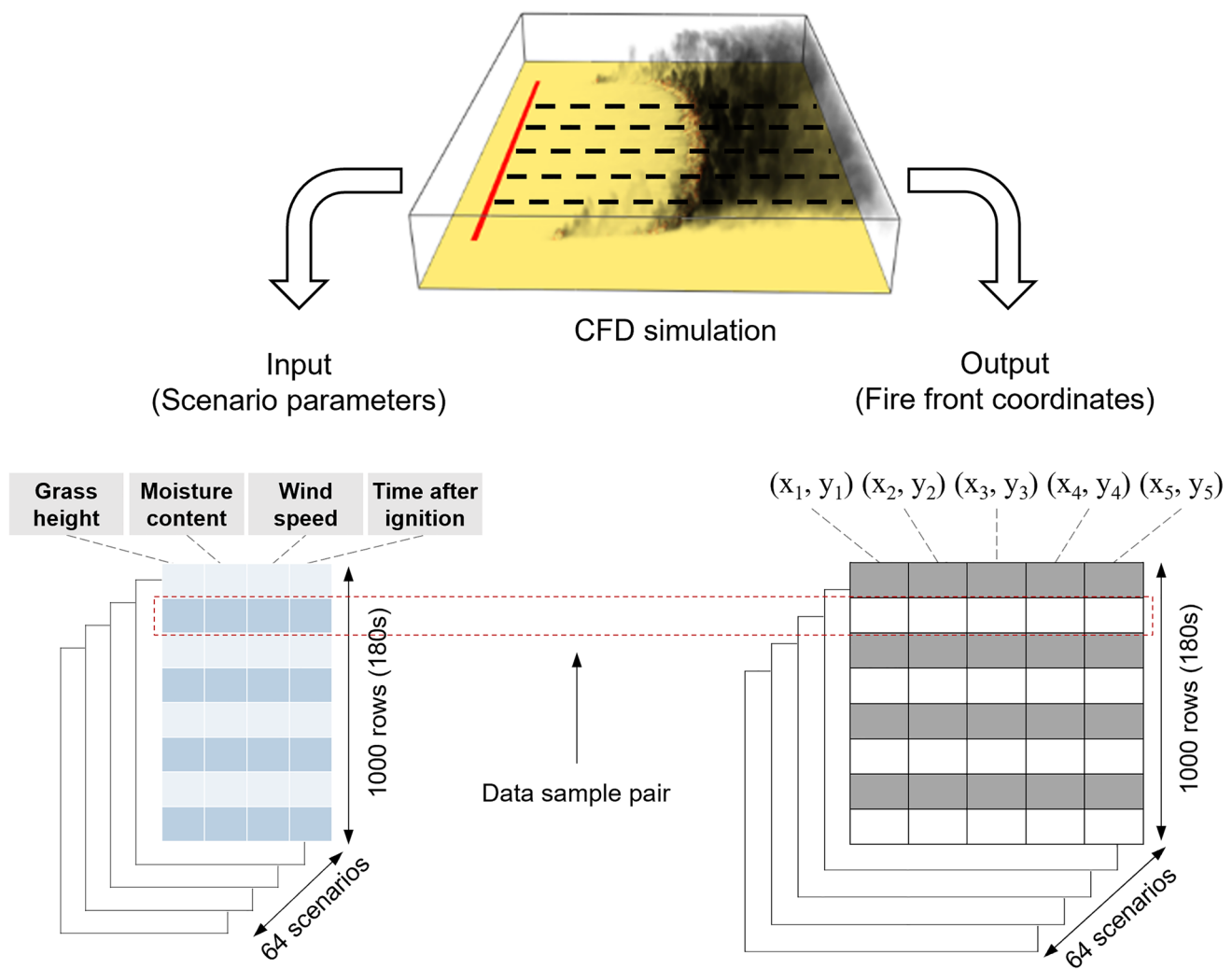


Fig. 5 Classification of the input and output parameters generated from simulation used in the machine learning model. For output parameterization, we tracked the evolving fire front by recording

coordinates at five strategic locations (x_1, y_1) to (x_5, y_5) across multiple time steps. These points were strategically positioned to capture critical fire regions: one at the head fire, two along each flank

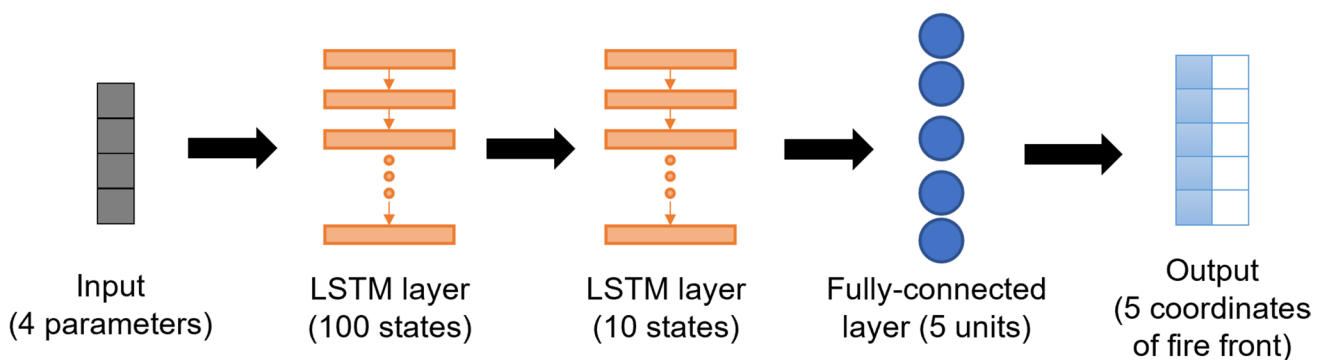


Fig. 6 Structure of the proposed LSTM network

the activation function, loss function, and optimiser, respectively. Then, the training dataset was used to fit the smodel for 500 epochs so that all the training samples were fed into the model 500 times. The selection of LSTM architecture and hyperparameters was informed by established work in the field (Islam et al. 2022; Li et al. 2025; Wu et al. 2022) and refined through empirical testing.

After training, the model performance on the training dataset (80% data) and validation dataset (20% data) during training are recorded. Figure 7 demonstrates the model performance evaluated by R^2 and loss (MSE) evolving with training epochs. The MSE quickly reaches a relatively low value and finally converges to around 0.01 on both training and validation datasets. The R^2 converges to around 0.96 after around 200 training epochs. The results indicate that the training epoch number of 200 is sufficient enough, and the proposed model well learns the relationship between the input and output.

Physics-informed LSTM with simulation and experiment data

In our initial study, we employed five key coordination points to efficiently represent the evolution of the firefront. This simplification preserved essential fire dynamics while ensuring computational tractability. To evaluate the generalizability of our LSTM model, we conducted six additional simulations with parameters detailed in Table 2.

To assess the model's robustness within the trained parameter range, we randomly selected six time points from these simulations and quantified the fire head deviation between predicted and simulated results using the following equation:

$$\text{HeadDeviation} = \frac{|\text{Prediction} - \text{Simulation}|}{\text{Simulation}} \quad (4)$$

The results demonstrate that the LSTM model effectively captures the underlying patterns of fire spread,

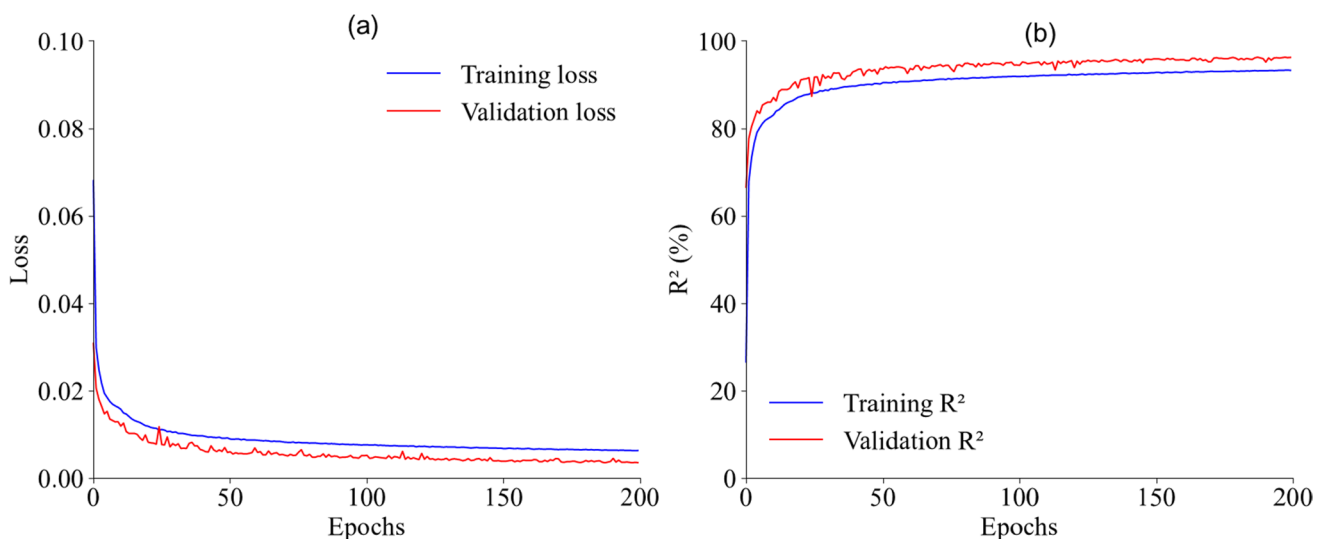


Fig. 7 The model performance **a** loss curve, and **b** R^2 curve

Table 2 Blind simulations to test the generalizability of LSTM model

No	Grass height	Moisture content	Wind speed	Random time	Head location in simulation/prediction
1	580	4.1	5.3	50	87.0/81.3
2	540	3.7	5.6	60	116.5/111.0
3	510	4.9	5.9	70	143.0/142.2
4	470	5.7	3.2	80	60.0/59.5
5	430	4.2	5.3	90	187.0/180.7
6	390*	5.3	4.6	100	179.5/179.0
Training range	420–600	4–7%	3–6		

*No.6 is an extrapolation case where 390 mm grass height is not within the training range

achieving strong agreement with the simulation data across all six blind tests. Notably, the average deviation was only 2.5%, indicating high predictive accuracy. Surprisingly, the model maintained stable performance even when tested with grass heights below the training range. This suggests a degree of extrapolation capability, highlighting the model's adaptability and efficiency in handling scenarios beyond its original training conditions.

Critically, our AI model was trained exclusively on synthetic data generated by FDS, with no exposure to real-world experimental measurements during the training phase. This strict segregation ensured that the model's predictive capabilities were rigorously tested against entirely independent data, eliminating any risk of data leakage or overfitting to experimental conditions.

Following the training of our Long Short-Term Memory network, we systematically validated its performance by comparing predictions with experimental observations from Case F19 (Cheney et al. 1993), a benchmark scenario characterized by a wind speed of 4.8 m/s, fuel moisture content of 5.8%, and grass height of 510 mm. The experimental dataset included firefront measurements at three distinct timesteps, enabling us to generate 15 predicted coordination points (5 points $\times$ 3 timesteps) for direct comparison with the empirical firelines, as illustrated in Fig. 8a. This validation approach, where a model trained purely on physics-based simulations is tested against independent physical experiments, underscores the generalizability of our hybrid methodology and its potential to bridge the gap between numerical modelling and real-world fire behaviour.

Our findings indicate that the AI model exhibits robust predictive capabilities, particularly in terms of accurately capturing the shape of the fire head and the rate of spread. Notably, during the initial 58 s, the forecasted results indicated a slower progression compared to the experimental data. However, this early discrepancy does not undermine the model's efficacy; instead, it highlights the adaptive nature of the LSTM approach. Furthermore, the model was trained along the five measuring lines, limiting its learning capability and significantly impacting the prediction at an early timestep.

The contours shown in Fig. 8a reveal notable patterns that warrant further explanation, particularly regarding the underprediction at early stages and the flat contour observed at later times. As highlighted in Fig. 8b, the initial underprediction can be attributed to limitations in the AI model's learning process. Specifically, during the early time steps, the model only has access to data from five measuring lines, with no observations at the flank positions, especially along measuring lines D and E. Consequently, the model observes a zero value for the fire front location during the first 35–45 s until a non-zero value is detected, as depicted in Figs. S4 and S5. This results in an inability to properly learn and spread the fire along the flanks in the initial stages. However, as time progresses, the model improves its learning, which leads to more accurate predictions of fire curvature, particularly around the 86-s time step.

As time continues, the model demonstrates its ability to recalibrate and align more closely with the baseline experimental data, suggesting that, although there are short-term variances in prediction speed, the model is capable

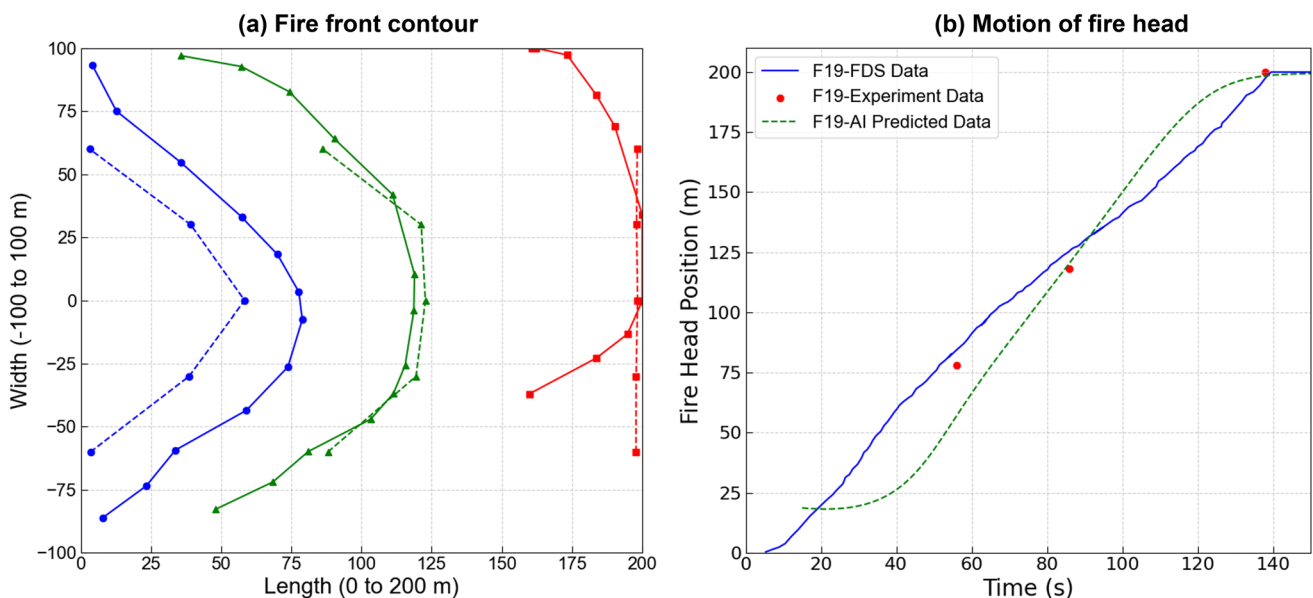


Fig. 8 **a** Fire line front contour comparison between experiment data (F19) of solid line and AI predictions of dashed line and **b** Fire head position comparison among experiment data (F19), CFD modelling, and AI predictions

of adjusting its forecasts to better reflect observed trends over time. This indicates that the AI model can eventually overcome initial limitations and provide reliable predictions, especially for the fire's head. At later stages, the flat contour observed in the fire's spread indicates a divergence in predictive accuracy between the head and the flanks. While the model demonstrates strong performance in predicting the fire head, the accuracy diminishes along the flanks. This discrepancy is primarily due to limitations in the training data, particularly for the flank firelines (B, C, D, and E). These firelines often exhibited abrupt interruptions in the experimental data, notably in the regions affected by wind vortices, as shown in Fig. 4a between $y = 50$ and 60 m. To address this, future work should prioritize expanding the dataset with targeted flank measurements, including high-resolution wind and curvature data to capture nonlinear propagation effects. Incorporating physics-based constraints (e.g., aerodynamic interactions) or hybrid modelling approaches could further improve flank predictions by accounting for empirically unresolved processes. Validation against controlled flank-spread experiments will be essential to assess these refinements.

These interruptions were particularly pronounced in the extreme flank data for lines D and E, as further illustrated in Figs. S4 and S5. Additionally, a shift in wind direction observed in the experimental setup (Cheney et al. 1993) was not considered in the current model, which could further explain the discrepancy in predictive performance. Similar challenges have been noted in previous studies, such as those by Vanella et al. 2021 with FDS version 6.7.5 and Mell et al. 2013 using earlier versions of FDS (WFDS). Both studies reported similar flat profiles at the end stage of their simulations. Innocent et al. 2023 also observed that variations in wind speed and direction during experiments, not adequately replicated in simulations, were a major contributing factor to this issue.

These findings have significant implications for the field of fire modelling and management. Improving the predictive accuracy of AI models in fire simulations could greatly enhance their applicability to real-world fire management scenarios, aiding in more effective decision-making for fire prevention and control strategies. Future research will focus on optimizing model parameters and incorporating additional environmental variables, such as changeable wind speed and direction, to enhance the model's overall performance and reliability.

Limitation and advantage discussion

The reliance on the F19 dataset for model validation, while providing a rigorous benchmark under controlled conditions, introduces inherent limitations concerning generalizability. Although the model exhibits strong agreement with the F19

dataset, its validation against a single experimental case raises concerns about its applicability to broader or more variable scenarios. Specifically, the F19 dataset's parameters may not fully capture extreme environmental conditions or highly heterogeneous fuel properties. For instance, while F19 includes key operational conditions—such as moderate wind speeds—it lacks representation of extreme scenarios (e.g., wind speeds exceeding 15 m/s) or spatially varied fuel distributions, which are critical for assessing model robustness in real-world applications.

Given these constraints, we recommend applying the model strictly within its validated parameter bounds (e.g., wind speeds of 4–7 m/s, moisture content of 3–6%, and uniform fuel distribution). Extrapolation beyond these ranges should be approached with caution, as predictive accuracy under untested conditions remains uncertain.

While the preceding discussion highlights the current limitations regarding model generalizability, it is important to contextualize the efficiency advantages of the proposed AI method relative to traditional approaches. As summarized in Table 3, the AI model demonstrates significant practical benefits in operational speed and resource requirements compared to the benchmark FDS simulations. Specifically, the AI method requires much less prediction time than FDS, and crucially, it does so without the computationally intensive simulation step required by FDS for each new scenario. This fast reaction capability stems from the upfront investment in training (2 h), which allows the AI model (LSTM model) to leverage learned patterns for rapid inference once deployed. Furthermore, the AI model achieves high accuracy (~95%) relative to the FDS benchmark within the validated F19 dataset conditions. However, it is essential to reiterate that these performance metrics, particularly the accuracy, are currently validated only within the specific bounds of the F19 dataset, as discussed in the limitations above. The computational efficiency represents a clear operational advantage for scenarios falling within the validated parameter space.

To mitigate these limitations, our team is currently compiling a multi-source dataset incorporating a wider range of environmental conditions and fuel properties. This expanded dataset will enable more comprehensive

Table 3 Training and Prediction cost for FDS and AI methods

Features	FDS simulation	LSTM AI method
Prediction delay for one case	10^3 s (slow modeling)	10^0 s (fast reaction)
Training time	N/A	1 h
Computational cost	Super High	Moderate
Knowledge	CFD modeling	LSTM model
Accuracy	Benchmark	~95%

validation, enhancing the model's reliability across diverse and complex scenarios.

Conclusions

The present study addresses the preliminary efforts involved in proposing a physics-informed LSTM model to estimate fire behaviour, specifically focusing on predicting fire head propagation. We first established a comprehensive numerical database derived from wildfire simulations conducted using the physics-based fire model to achieve this objective. This database served as a synthetic dataset for training our deep learning model, providing a robust foundation for subsequent analyses.

By leveraging the strengths of both physics-based modelling and deep learning, we aim to enhance the accuracy and reliability of fire spread predictions. To validate the effectiveness of the physics-informed LSTM model, we validated our model involving independent numerical cases alongside experimental data. This rigorous validation process allows us to quantify the model's efficacy and ensure its applicability to real-world scenarios. Through these comparisons, we assess the model's performance in accurately predicting fire behaviour under various conditions, thereby contributing valuable insights to the field of fire modelling and management.

Our findings underscore the generalizability of our hybrid methodology and its potential to bridge the gap between numerical modelling and real-world fire behaviour, demonstrating substantial potential for applications in fire risk assessment and mitigation strategies. While the model exhibits strong predictive accuracy for the fire head, its performance in representing the flanks is less precise. This discrepancy, consistent with observations in prior research, is primarily attributed to variations in wind speed and direction during experiments that were not fully replicated in the simulations. Such variability in environmental conditions significantly influences lateral fire spread, highlighting a critical area for improvement. To address these limitations, future work will focus on refining the model, expanding the dataset, and incorporating additional factors influencing fire behaviour to enhance overall predictive capabilities.

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Investigation, Methodology, Writing – original draft. Duncan Sutherland: Methodology, Writing – review & editing. Khalid Moinuddin: Resources, Writing – review & editing. Xinyan Huang: Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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